

The Lebesgue measure on an open interval in $\mathbb{R}$
 $(a, b) \subseteq \mathbb{R}$ is defined (with $a < b$)

$$\mu((a, b)) = b - a$$

- a) Show that $\mu(\{a\}) = 0 \forall a \in \mathbb{R}$

From the definition of a measure we have

$$0 \leq \mu(\{a\})$$

Also $0 < \mu(\{a\})$

Choose $b, c \in \mathbb{R}$ such that $a \in (b, c)$

From the definition of the measure on the open interval we have

$$\mu((b, c)) = c - b$$

But we can write the interval as a union of disjoint sets:

$$(b, c) = (b, a) \cup \{a\} \cup (a, c)$$

Since μ is a measure

$$\begin{aligned} \mu((b, c)) &= \mu((b, a) \cup \{a\} \cup (a, c)) = \mu((b, a)) + \mu(\{a\}) + \mu((a, c)) \\ &= (a - b) + \mu(\{a\}) + (c - a) = c - b + \mu(\{a\}) \end{aligned}$$

But this implies

$$\mu((b, c)) = c - b < c - b + \mu(\{a\}) = \mu((b, c))$$

Which is of course a contradiction

Thus we must have $\mu(\{a\}) = 0$

- b) Show that $\mu((a, b)) = \mu([a, b)) = \mu((a, b]) = \mu([a, b]) = b - a$

Since these are all the unions of the open interval and the end points which all have measure 0 this is true

$$\begin{aligned} \mu([a, b)) &= \mu(\{a\}) + \mu((a, b)) = 0 + b - a \\ \mu((a, b]) &= \mu(\{b\}) + \mu((a, b)) = 0 + b - a \\ \mu([a, b]) &= \mu(\{a\}) + \mu(\{b\}) + \mu((a, b)) = b - a \\ \mu((a, b)) &= 0 + b - a \end{aligned}$$

Here all the measures of the points in $\mathbb{R}$ are 0 from (a)

- c) Show that $\mu((a, \infty)) = \mu([a, \infty)) = \mu((-\infty, a)) = \mu((-\infty, a]) = \infty$

First of all we notice

$$\begin{aligned} (a, \infty) &\subset [a, \infty) \Rightarrow \mu((a, \infty)) \leq \mu([a, \infty)) \\ (-\infty, a) &\subset (-\infty, a] \Rightarrow \mu((-\infty, a)) \leq \mu((-\infty, a]) \end{aligned}$$

We write the interval as a countable union of disjoint intervals and use properties of the measure to compute the whole

$$(-\infty, a) = \bigcup_{n \in \mathbb{N}} [a - n, a + 1 - n)$$

These disjoint subsets all have measure 1 giving us

$$\mu((-\infty, a)) = \sum_{n \in \mathbb{N}} \mu([a - n, a + 1 - n)) = \infty$$

Also:

$$\mu((-\infty, a)) \leq \mu((-\infty, a]) \Rightarrow \mu((-\infty, a]) = \infty$$

The others are the exact same approach

$$(a, \infty) = \bigcup_{n \in \mathbb{N}} (a + n - 1, a + n]$$

$$\mu((a, \infty)) = \sum_{n \in \mathbb{N}} \mu((a + n - 1, a + n]) = \infty$$

And

$$\mu((a, \infty)) \leq \mu([a, \infty)) \Rightarrow \mu([a, \infty)) = \infty$$

And were done 😊